

Separating State Loss from Planning Error in Long-Horizon Transformer Reasoning

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Abstract

Language models degrade as reasoning chains lengthen. The usual remedies act at inference time, through tree search, process supervision, or repeated sampling, and they treat the underlying policy as a fixed object whose failure is never diagnosed. We argue the obstacle is measurement. On open-ended text there is no ground-truth intermediate state, so the accuracy drop reported by long-context benchmarks confounds two different failures: the model has lost track of the world, or the model holds the world correctly and chooses badly. This paper separates them. We study reasoning in domains carrying an exact, cheap, external state oracle: chess played from move notation with no search at inference, and a synthetic variable-tracking task whose answers require computing over the tracked state. Linear probes read the residual stream against the oracle at every token, which yields state-tracking error and, conditioned on a verified state, planning error. On 120,000 games we measure how far a model reasons before it loses the position, reported against majority, label-permutation, and randomised-weight controls. We then evaluate a small gated recurrent side-channel of fixed width, trained with an auxiliary state-reconstruction objective, which amortises state instead of re-deriving it, and we test whether any gain localises to state tracking rather than to judgement. We report a failed prediction and a revised metric openly, and we state plainly what our compute budget left untested.

Keywords: long-horizon reasoning, state tracking, Transformer memory, probing, recurrent bottleneck, design science

1 Introduction

A model that reasons over many steps has to keep track of something. A Transformer stores nothing between positions. Each token’s representation is rebuilt from the whole prefix through attention (Vaswani et al., 2017), so any quantity the task requires, a board position, the current value of a variable, which sub-goal is still open, exists only because the network re-derives it, at every layer and every position, from the beginning. That re-derivation is dependable over short spans and unreliable over long ones. The symptom is familiar to anyone who has watched a model reason for forty steps: it stays fluent, and it quietly stops being right about its own earlier commitments.

The responses in common use sit outside the model. Tree search grafts a deliberation procedure onto the policy (Yao et al., 2023). Process reward models supervise intermediate steps with a learned verifier (Lightman et al., 2024). Self-consistency samples several chains and votes (Wang et al., 2023). Each raises end-task accuracy, and each leaves the original question untouched: what exactly fails inside the network as the chain lengthens? Answering that means reading an intermediate representation against a known-correct intermediate state. On open-ended text no such state exists. The intermediate “state” of an argument in prose is not well defined, which is why evaluation here is noisy and why architectural claims are hard to falsify.

The consequence reaches past research practice. A person working through a discharge summary, a tenancy agreement, or a benefits calculation with a model’s help is rarely positioned to audit step twenty-eight. When the state has been lost, the output still reads as competent. The failure is invisible precisely to the reader who most needs to see it, which is a strong argument for measuring it rather than covering it with more inference-time compute.

Our position. Study the failure where state is exact, cheap, and external. Chess presented purely as move notation, modelled by a standard autoregressive decoder with no tree search at inference, offers what prose does not:

- The true state after any prefix is computable by a rules engine, supplying a per-token label for probing.
- Legality is decidable, so state-tracking failure has an unambiguous observable. An illegal move proves the position was lost. It is not a matter of taste.
- Move quality is separately scoreable by an external engine, so planning failure can be measured conditional on the state being intact.
- Depth is a controlled variable. Ply index is uniformly scaled, unlike “number of reasoning steps” in prose.
- The domain is combinatorial and adversarial, so surface heuristics do not substitute for computation.

Chess is our instrument, not our subject. We use it to measure a failure mode at a signal-to-noise ratio text does not allow, then repeat the measurement on a second domain to test whether what we found is a property of long-horizon state tracking or a property of chess.

Research questions and objectives. Each question is paired with an objective, and each objective names the section discharging it.

- **RQ1.** Can the accuracy loss of long-horizon reasoning be separated into state-tracking error and planning error? **RO1:** define both quantities and give an estimator for each (Sections 4 and 6).
- **RQ2.** How far does a search-free autoregressive model reason before its internal state stops being recoverable, and how does that depth respond to more layers and more width? **RO2:** measure the horizon across a model grid against trivial baselines (Section 9).
- **RQ3.** Does a fixed-width recurrent channel, supervised to carry state, extend that depth, and does any gain localise to state tracking rather than judgement? **RO3:** evaluate the artifact with parameter-matched and ablated controls (Sections 7 and 9).
- **RQ4.** Does any of this survive a change of domain? **RO4:** repeat the decomposition on a second oracle-bearing task (Section 8).

Contributions.

1. A decomposition of long-horizon error into two separately estimable quantities, where existing long-context evaluation reports only their sum.
2. A characterisation of the planning horizon, and of the mismatch we call the state-space bottleneck: attention pays memory linear in context length to re-derive a state whose description length is constant.
3. An artifact, the Adaptive Latent State Bottleneck, a gated recurrent side-channel of fixed width with an auxiliary state-reconstruction loss, together with the ablations needed to attribute any gain to it.

4. A second oracle-bearing domain, so findings obtained with the chess instrument can be checked against a task that is arithmetic rather than spatial.
5. Transferable design knowledge, stated as design principles, for measuring long-horizon reliability wherever an external state oracle can be constructed.

We frame the work as design science in the sense of Hevner et al. (2004) and Peffers et al. (2007), where the contribution is a justified artifact plus the design knowledge it carries (Gregor and Hevner, 2013), evaluated against a real problem rather than against a single accuracy number.

2 Related Work

Inference-time search over language models. Deliberation over candidate continuations (Yao et al., 2023), process supervision (Lightman et al., 2024), and marginalising over sampled chains (Wang et al., 2023) all buy accuracy with inference compute, built on chain-of-thought prompting (Wei et al., 2022). They take the base policy as given. Our concern is the base policy: a search wrapper spending its budget correcting state-tracking errors compensates for a representational defect, and the compensation cost scales with the defect. What this paper does differently is diagnose the defect instead of budgeting around it.

Long-context degradation. Position-dependent retrieval failure (Liu et al., 2024) and effective-context measurement (Hsieh et al., 2024) establish that usable context is shorter than advertised context. These measure retrieval of material present in the prefix. Long-horizon reasoning failure differs in kind: the needed quantity was never written down, only implied, and must be maintained through composition. A model can attend perfectly to every token of a sixty-ply game and still not know where the pieces are. What this paper does differently is measure maintenance rather than retrieval.

State tracking and probing. Linear probes recovering board state from game-transcript models (Li et al., 2023), the linear structure of those representations (Nanda et al., 2023), chess used explicitly as a state-tracking testbed (Toshniwal et al., 2022), and causal editing of internal associations (Meng et al., 2022) together establish that such models form internal world models and that intervening on them changes behaviour. We take that as our starting point rather than our result. Given that a state representation exists, we ask how its fidelity decays with depth, whether the decay explains the behavioural failure, and whether an architectural prior arrests it. Probe design follows the control-task discipline of Hewitt and Liang (2019). What this paper does differently is treat decodability as a function of reasoning depth rather than as a single number.

Expressivity of state. Theoretical work bounds what these architectures can track: state-space models are more limited than their recurrent form suggests (Merrill et al., 2024), and intermediate tokens change what a Transformer can express (Merrill and Sabharwal, 2024). That literature asks what is representable in principle. We ask what is maintained in practice by a trained model, and at what depth it stops.

Recurrence and state-space models. Structured state-space models (Gu et al., 2022) and selective variants (Gu and Dao, 2024) provide constant per-token state, and hybrid stacks are ordinary practice. The gains usually reported are throughput and context length. Our claim is narrower and more testable: a small recurrent channel, explicitly supervised to carry task state,

Table 1: Design principles and their realisation.

Design principle	Realisation in this work
DP1. Ground the diagnostic in an external oracle, not in model self-report.	A rules engine labels the exact state after every token, so probe targets are independent of the model being measured.
DP2. Separate the two failure modes before attributing any effect.	Planning error is scored only where the top-1 move is legal and the state is independently verified as intact.
DP3. Establish what a trivial predictor achieves before claiming a representation exists.	Every decodability number is reported against a per-square majority predictor, a label-permutation control, and a randomised-weight model of identical shape.
DP4. Amortise state rather than re-deriving it at every position.	A gated recurrent channel of fixed width carries the realised state forward and is read back into the residual stream.
DP5. Attribute a gain, do not merely observe one.	Architecture, auxiliary supervision, and read-out are varied independently, against a parameter-matched attention-only control.

improves long-horizon correctness at negligible parameter cost. That is a different proposition from efficiency.

Search-free chess models. Policies reaching strong play by distilling engine evaluations without a tree (Ruoss et al., 2024) show the task is learnable without search. Their objective is playing strength. Ours is measurement, so we deliberately work with models weak enough to fail, because a model that never loses the position teaches nothing about losing the position.

3 Design Problem and Requirements

The problem, stated plainly: long-horizon reasoning failure is reported as a single number, so it cannot be acted on. A practitioner seeing accuracy fall at step thirty cannot tell whether to buy a bigger model, add a verifier, or change the architecture, because the measurement does not distinguish the causes. The artifact we build is a measurement procedure first and an architectural component second.

In scope. Search-free autoregressive decoders; domains with an exact, cheaply computable state oracle; state of bounded description length.

Out of scope. Inference-time search procedures, which we neither use nor evaluate; tasks whose state grows without bound, such as open-ended fact accumulation; and any claim about production-scale models, which our compute budget does not license.

4 Problem Formalisation

Let a task be a token sequence $x_{1:L}$ generated left to right, and let $s_t = \Phi(x_{1:t})$ be the exact state after prefix $x_{1:t}$, computed by an external oracle Φ . For chess the state is piece placement, side to move, castling rights, and en-passant target. The state space is large, but the description length of any s_t is constant in t : a position occupies a few hundred bits however many plies produced it.

Let $h_t^{(\ell)} \in \mathbb{R}^d$ be the residual-stream activation at position t and layer ℓ , and let $g^{(\ell)}$ be a linear decoder trained to recover s_t from it.

Definition 1 (State fidelity). $F^{(\ell)}(t) = \Pr[g^{(\ell)}(h_t^{(\ell)}) = s_t]$, the probability the exact state is linearly decodable at depth t and layer ℓ .

Definition 2 (Error decomposition). *State-tracking error* $E_{\text{state}}(t) = \Pr[\hat{s}_t \neq s_t]$, estimated by the illegal-move rate and by probe-oracle disagreement. *Planning error* $E_{\text{plan}}(t) = \Pr[\text{move is a blunder} \mid \hat{s}_t = s_t]$, estimated by engine centipawn loss conditional on an intact state. *Observed error* is $E_{\text{state}} + (1 - E_{\text{state}})E_{\text{plan}}$, and existing benchmarks report that sum alone.

This decomposition is the methodological core of the paper. An intervention lowering the sum tells us little. An intervention lowering E_{state} specifically, and thereby extending the depth at which state survives, is evidence about memory rather than about taste.

5 The State-Space Bottleneck

Proposition 1 (Cost mismatch). *For an attention-only decoder, computing $h_t^{(\ell)}$ attends over t prior positions, so maintaining state costs $O(t)$ reads and $O(t)$ key-value memory, while the state itself has description length $O(1)$. Nothing in the architecture amortises it: s_t and s_{t+1} differ by one move and are computed independently.*

Two consequences follow, and we tested both. First, fidelity should fall with t even well inside the context window, because the re-derivation circuit must compose more update steps within a fixed layer budget. Depth of composition, not distance of retrieval, is the binding constraint. This is supported (Section 9).

Second, we predicted fidelity would peak at intermediate layers and fall toward the output as representations specialise for next-token prediction, the state being a means rather than an end. **That prediction is wrong.** State decodability rises through the stack and is maximal at the final block, moving from ?? at the first layer to ?? at the last. The reading we now prefer is that for this task the state sits close to the target: the next legal move is a direct function of the position, so a representation making the position linearly available is also a good representation for the language-modelling head. We keep the failed prediction because it bears on where a state carrier should be inserted, and our spacing was chosen before we knew this.

6 Diagnostic Protocol

Data. Public game records filtered to a rating floor, deduplicated by move-sequence hash, and split by *game* rather than by position, so no prefix is shared between the probe-fitting split and the evaluation split. The corpus holds 120,000 games: 96,000 for training, 12,000 for probe fitting, and 12,000 for evaluation, with a mean length of 78.2 plies and a move vocabulary of 1,926.

Oracle. A rules engine replays every game and emits the exact state after every token. Replay is verified complete before any model is measured.

Probes. One linear head per layer, fitted on the probe split and evaluated on the held-out split. Because most squares are empty, raw per-square accuracy is misleadingly high, so we report accuracy restricted to occupied squares and compare against a per-square, per-depth majority predictor.

Controls. Three, following DP3. A majority predictor; a label-permutation control pairing activations with another game’s state at the same depth, which preserves label marginals while destroying the relation to the input; and a randomised-weight model of identical architecture, bounding what probe capacity alone extracts.

Behavioural measures. Illegal top-1 move rate against depth, and the probability mass the model assigns to legal continuations. Both are read against the oracle position, so they measure the model and not the probe.

Causal check. We edit probe-identified state directions toward a counterfactual occupancy and ask whether the move distribution shifts accordingly. A representation merely correlated with the state would not respond.

7 Adaptive Latent State Bottleneck

After every k -th attention block we insert a recurrent channel carrying $z_t \in \mathbb{R}^{d_s}$ with $d_s \ll d$:

$$u_t = W_{\text{in}} h_t, \quad \alpha_t = \sigma(W_g[h_t; z_{t-1}]), \quad z_t = (1 - \alpha_t) \odot A z_{t-1} + \alpha_t \odot \phi(u_t),$$

read back as $h_t \leftarrow h_t + W_{\text{out}} z_t$. The transition A is diagonal, so inference-time state is $O(1)$ per token. The gate is the adaptive element: writing is cheap when a move perturbs the state locally and expensive when it does not, so capacity is spent on state change rather than uniformly across tokens. Training adds an auxiliary term, $\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda \mathcal{L}_{\text{state}}$, supervising z_t to be the state. The oracle is not needed at inference.

Following DP5 we separate three things that are easily confused. The architecture alone is $\lambda = 0$. The supervised channel is $\lambda > 0$. The contribution of the read-out is isolated by zeroing W_{out} at test time. A parameter-matched attention-only control, with a widened feed-forward block absorbing the same parameter count, guards against reading a capacity effect as a structural one.

8 Domain 2: variable-state tracking

Chess is one domain, and a reader is entitled to ask whether anything measured with it is a fact about long-horizon reasoning or a fact about chess. We repeat the decomposition on a second task sharing only the property we depend on, an exact per-token oracle.

A program updates four integer variables modulo twenty over five to thirty-two steps, each step of the form $v_i \leftarrow v_j \oplus c$, then queries $(v_a + v_b) \bmod 20$. The state is arithmetic rather than spatial, updates are sparse, and the answer needs a computation on top of the retrieved state, so the conditioning defining planning error is exact rather than estimated: we know which two variables the answer depends on and can ask whether both are decodable at the query token. Depth is the number of steps.

An earlier version used eight variables modulo one hundred. It sat exactly at chance, and we report that because it shaped the design: a task no condition can learn measures nothing, so we reduced it until degradation was visible inside our budget, for the same reason the chess models are deliberately small.

This is not the natural-language mathematics test our protocol specifies (Cobbe et al., 2021; Hendrycks et al., 2021), and we do not present it as one. It answers the weaker question of whether the intervention generalises past the instrument it was designed against.

Table 2: Chess arm. Occupied-square state accuracy (occ), illegal top-1 move rate, and behavioural horizon H_{ill} (first ply bucket where the illegal rate exceeds the threshold). Mean $\pm$ s.d. over seeds.

Model	Params	seeds	occ@40–50	illegal@40–50	$H_{\text{ill}}(.05)$	$H_{\text{ill}}(.10)$
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9 Evaluation

9.1 Pre-registered predictions

H1 Fidelity declines with depth for attention-only models, with the horizon well inside the context window.

H2 The horizon scales weakly with depth and width, so scaling at fixed compute does not substitute for a state carrier.

H3 The artifact extends the horizon and lowers E_{state} at matched parameters and matched tokens.

H4 The improvement concentrates in E_{state} and not in E_{plan} .

H5 Gains carry to another long-horizon state-tracking domain.

H4 is the hypothesis that can end the paper. If the artifact improves everything uniformly, the mechanism we claim is wrong even when end-task numbers improve, and we would report that.

9.2 Results

9.3 A revised metric, declared

Our horizon was pre-registered on exact-position fidelity. On these models that quantity reaches zero by roughly ply ?? in every condition, so it cannot discriminate between them. We therefore also report horizons defined on occupied-square accuracy and on the illegal-move rate. The revision was made after inspecting baseline data and before evaluating any artifact condition, and all definitions are applied unchanged across conditions. We flag it rather than presenting the revised metric as the original plan.

10 Discussion

The decomposition is what we would keep if we had to discard everything else. It turns a single degradation curve into two curves with different remedies. Where state-tracking error dominates, more inference-time search is an expensive way to buy back what the architecture gave away, and a cheap structural change is the better lever. Where planning error dominates, the opposite holds. Reporting only the sum leaves a practitioner unable to tell which world they occupy, and the two worlds recommend opposite purchases.

The measurement discipline generalises past our two domains. Any setting where an external checker can reconstruct the intended intermediate state, program execution traces, spreadsheet recalculation, structured database updates, ledger or inventory state, supports the same protocol. The requirement is not chess. It is an oracle.

We also think the honest negative results here are part of the contribution. A predicted layer profile turned out to be backwards, a pre-registered metric turned out to be degenerate, and a first version of the second domain turned out to be unlearnable. Each of these shaped the design, and each is reported in the place where it changed a decision.

11 Implications for Individuals, Organisations, and Society

For the people who rely on long answers. Someone working through a discharge summary, a tenancy agreement, or a benefits calculation with a model’s help is rarely able to audit step twenty-eight. When state is lost the output stays fluent and stops being true, and fluency is exactly the signal a non-expert reader uses to decide whether to trust it. A diagnostic reporting where a model stops holding the thread gives that reader something currently unavailable: a stated range within which the tool is dependable. This matters most for people without an expert to check the work, which includes older users, sole traders, small clinics, and students.

For managers and organisations. Long-horizon reliability is a property to be governed, not a number to be procured. A team measuring only end-task accuracy cannot tell whether to budget for inference-time search, for a larger model, or for an architectural change, and these carry very different cost structures: search is a recurring per-query cost, a structural fix is paid once. The decomposition supports ordinary management practice: triage of which workloads exceed a measured horizon, human review placed at the depth where reliability falls rather than spread uniformly, and audit evidence that a limit was known and respected.

Strategic implications. An organisation able to measure its own horizon stands in a stronger position than one accepting a vendor’s benchmark average. The diagnostic here needs a rules engine and a probe, not privileged access to model internals at scale, so it can be owned and audited in-house rather than bought. Treating a measured reliability limit as documentation rather than as an embarrassment turns a weakness into a defensible position with auditors and regulators.

For society and digital equity. Paying for reliability with inference-time compute concentrates capability among those who can afford to spend it on every query. A structural fix costing a small fraction of parameters, if it holds, is the kind of remedy that travels to low-resource deployments, on-device assistants, and public-sector systems that cannot buy deliberation for every request. The equity question is not whether frontier systems can reason for fifty steps. It is whether affordable ones can be honest about the point at which they stop.

12 Limitations

- L1 Scale.** Our models are small and trained on a modest corpus, far below both the chess-specific literature and any frontier system. Undertrained models are easier to study because they fail within a measurable range, which is why the instrument works, but the horizon values we report describe these models and not larger ones.
- L2 The natural-language transfer test was not run.** It needs two matched general-corpus pre-trainings, which our budget did not cover. The experiment is absent rather than approximated, and the synthetic domain is not a substitute for it.
- L3 Oracle dependence.** Auxiliary state supervision needs an oracle during training. It exists for chess and for synthetic programs and does not exist for general corpora, so the unsupervised condition is the one that would carry any practical application.
- L4 Fixed-width state.** A constant channel width is a hard capacity ceiling, and tasks whose state grows without bound should not benefit.
- L5 Probes show presence, not use.** Linear decodability shows information is available, not that the model reads it. The patching experiment bounds this concern without eliminating

it.

L6 A metric was revised after seeing data. Declared in Section 9, applied uniformly, and reported alongside the original.

L7 Both domains are constructed. They carry exact oracles by design, which is what makes them measurable and also what makes them unrepresentative of open-ended reasoning.

13 Avenues of Future Research

FR1 Repeat the horizon measurement across a wider scale ladder, so the relationship between model size and horizon is estimated rather than bracketed by two points.

FR2 Run the natural-language mathematics transfer test with matched pre-training, which is the experiment deciding whether this line of work has practical reach.

FR3 Develop state supervision that does not need an oracle, for example by deriving approximate targets from execution traces or from consistency between sampled continuations.

FR4 Study adaptive or growing state width for tasks whose state is not of bounded description length.

FR5 Strengthen the causal analysis beyond single-direction patching, toward interventions establishing which computations consume the state representation.

FR6 Pre-register horizon metrics on pilot data, so the metric selection we had to declare here is settled before confirmatory runs.

FR7 Build oracle-bearing diagnostics for domains not designed as benchmarks, such as program execution or ledger reconciliation, to test whether the protocol survives contact with tasks we did not construct.

14 Conclusion

Long-horizon failure in Transformers is usually measured as one number and treated with more inference-time compute. We have argued it should first be measured as two, state lost against plan chosen badly, and that doing so needs a domain where an external oracle can say what the model should have been holding. Chess supplies one, a synthetic variable-tracking task supplies another, and the protocol needs only that an oracle exists. The people who stand to gain are not primarily the builders of frontier systems. They are the readers who cannot check step twenty-eight, and the organisations that have to decide, with evidence, how far to let a model reason before a person looks.

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